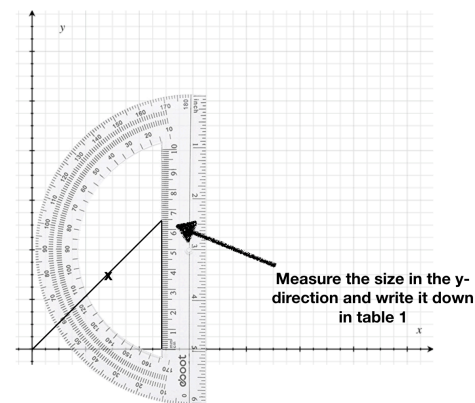
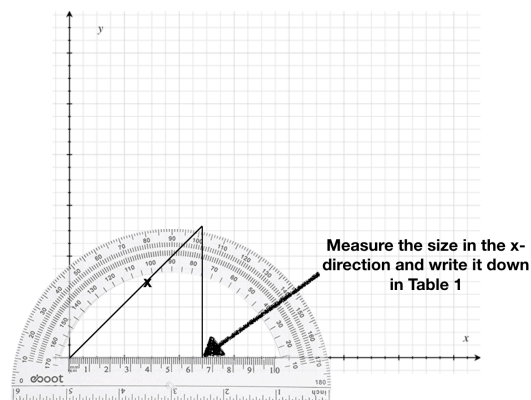
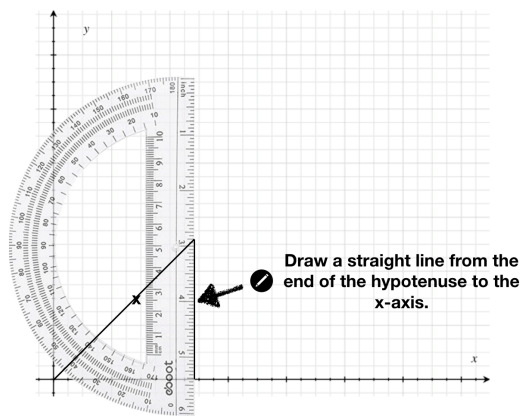
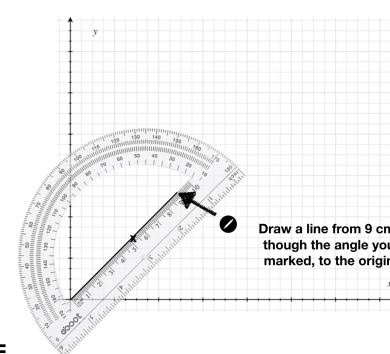
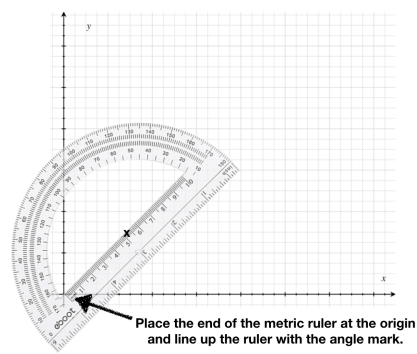
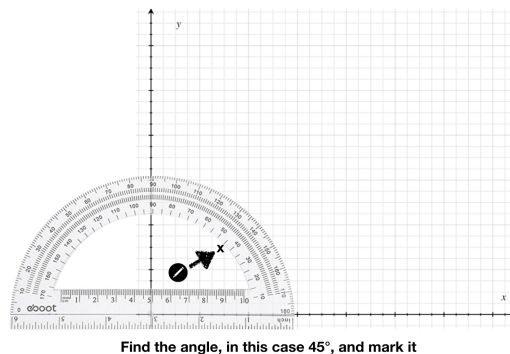
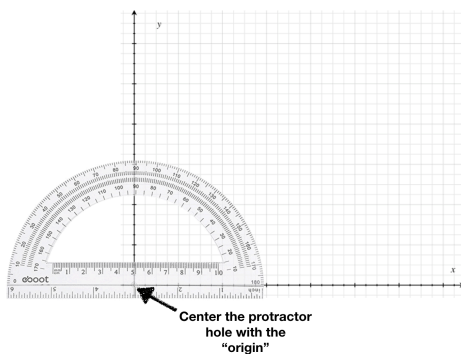


Trigonometry Worksheet

Using your protractor, draw 5 “triangles” all with the same hypotenuse length of $r = 9.0$ cm on each figure provided **after this page**. Each figure notes the angle to be used for the triangle. Note that 0° is a horizontal line (x) and 90° is a vertical line (y). Measure to the nearest tenth of a centimeter (e.g, 8.8 cm). **Work with a partner.**

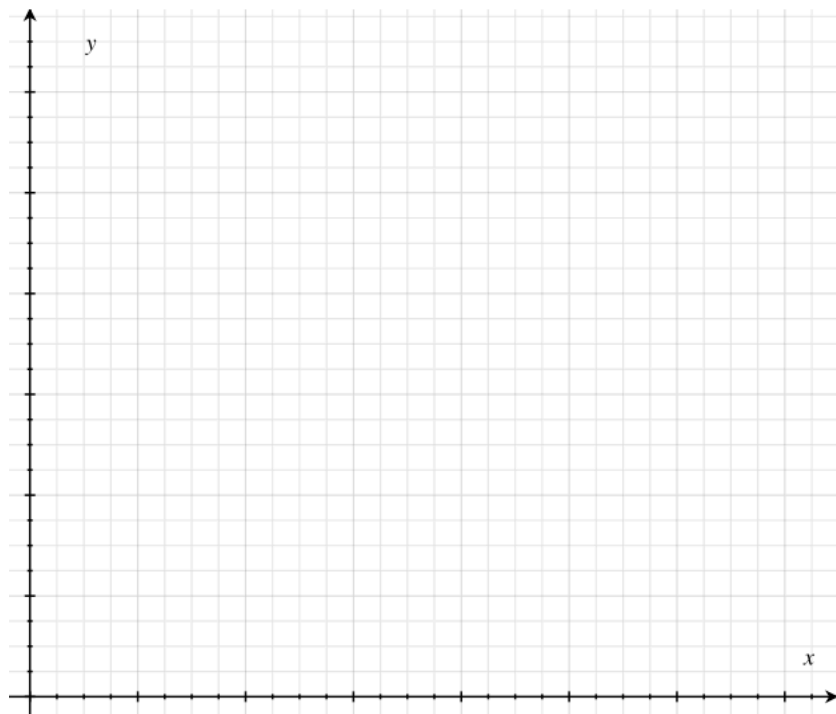
Example of how to draw a triangle with your protractor:

Suppose you are drawing a triangle with an angle (θ) of 45° ...

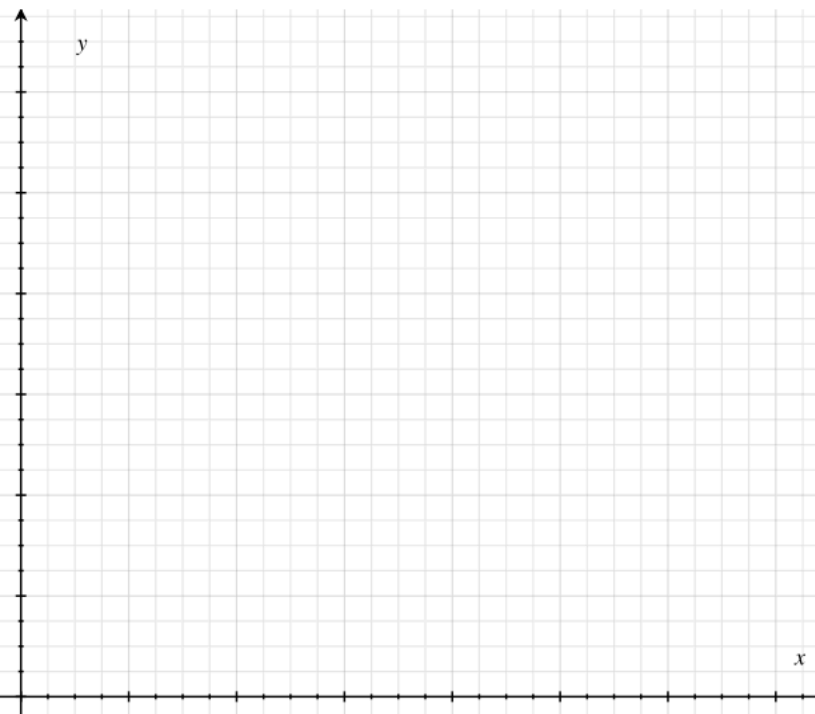


Now let's draw and measure some triangles...

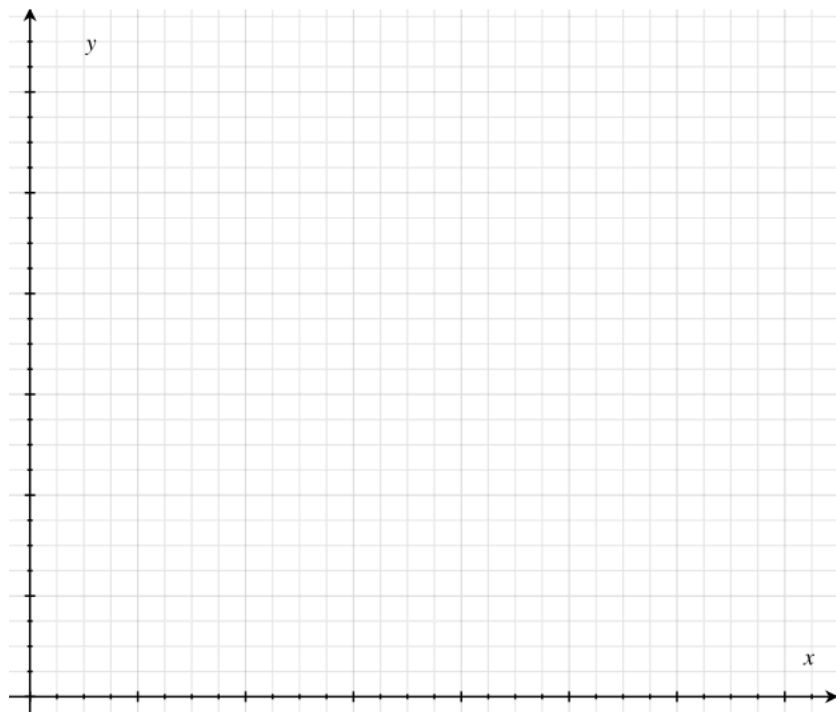
$\theta = 0^\circ$



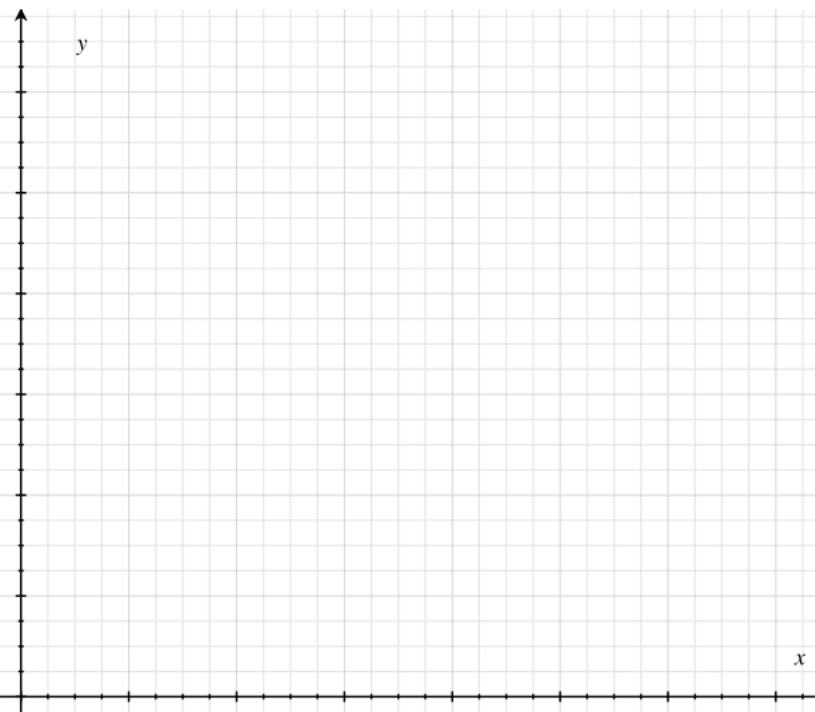
$\theta = 20^\circ$



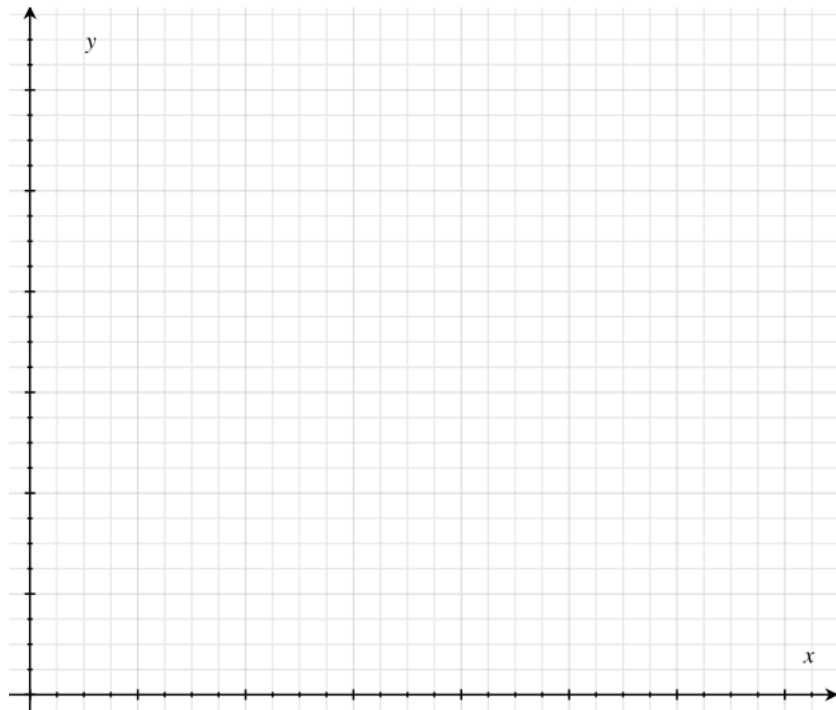
$\theta = 40^\circ$



$\theta = 60^\circ$



$\theta = 90^\circ$



Save this graph for later ($\theta = 40^\circ$)

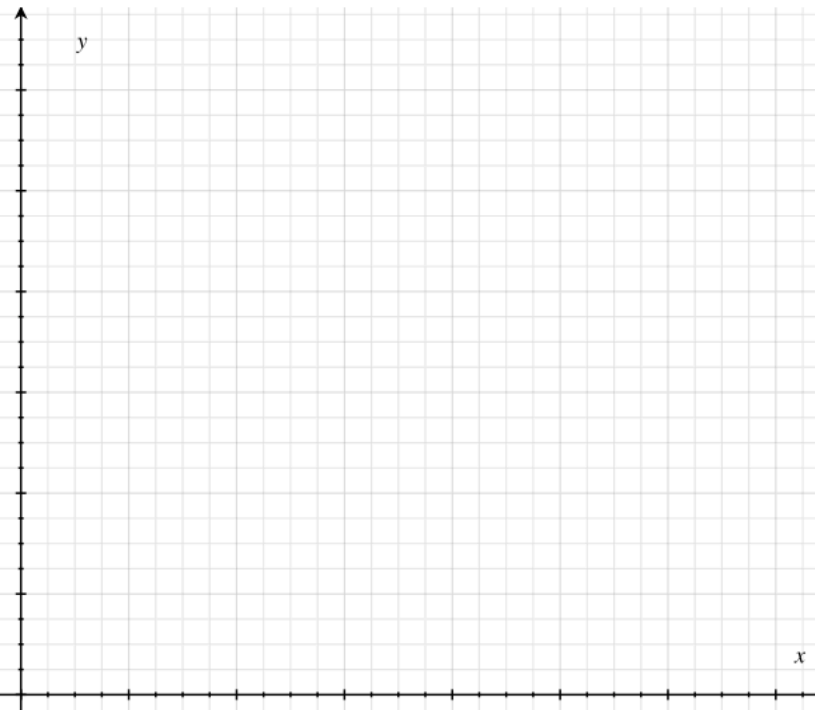


TABLE 1

Measure the x and y lengths of each triangle. Label the r , θ , x , and y on each triangle and then fill in the table below. Verify that $r = \sqrt{x^2 + y^2}$ before you fill in the other calculated values. For each measured value, write down the value to the nearest tenth of a centimeter (0.1). For each calculated value, write down the value to the nearest hundredth of a centimeter (0.01). The first row is filled out for you.

Hypotenuse length:

$r = 9.00$ cm

θ	x [cm]	y [cm]	$\sqrt{x^2 + y^2}$	x/r	y/r	y/x
0°	9.0	0.0	9.00	1.00	0.00	0.00
20°						
40°						
60°						
90°						

Why is y/x blacked out for 90° ?

TABLE 2

Calculate the following trig functions to the nearest 0.01 using a calculator. The first row is filled out for you. Note than some calculators (e.g., Google) use the function name “arctan” instead of $\tan^{-1}$. **Don’t forget to set your calculator to use degrees!**

θ	$\cos(\theta)$	$\sin(\theta)$	$\tan(\theta)$	$\tan^{-1}(y/x)$
0°	1.00	0.00	0.00	0.00
20°				
40°				
60°				
90°				

Compare the columns in Table 1 to the columns in Table 2. Complete the following equations using this comparison. The first equation is filled in for you. **Note that the values you calculated in the tables above are not perfectly precise!**

$r = \sqrt{x^2 + y^2}$
 $\cos(\theta) =$
 $\sin(\theta) =$
 $\tan(\theta) =$
 $\tan^{-1}(y/x) =$
 $\tan^{-1}(\tan(\theta)) =$

Choose a new value of r and draw another triangle at 40° in the graph we were saving for later. Measure x and y and then calculate all of the same values in table 1 and 2. Compare to the values in the 40° row for $r = 9$ cm.

$\theta = 40^\circ$

r	x [cm]	y [cm]	$\sqrt{x^2 + y^2}$	x/r	y/r	y/x	$\cos(\theta)$	$\sin(\theta)$	$\tan(\theta)$	$\tan^{-1}(y/x)$

Using the two trigonometric identities presented below, express r in terms of x , y , and/or θ (i.e. solve for r):

$\cos(\theta) = x/r$
 $r =$

 $\sin(\theta) = y/r$
 $r =$